Unit 1: Introduction to Machine Learning & Artificial Intelligence

Introduction to Artificial Intelligence (AI)

Artificial Intelligence (AI) is the simulation of human intelligence processes by machines, especially computer systems. These processes include learning (the acquisition of information and rules for using the information), reasoning (using rules to reach approximate or definite conclusions), and self-correction.

Key areas of AI include:

- Natural Language Processing (NLP)
- Computer Vision
- Robotics
- Expert Systems

Real-world examples of AI:

- Siri and Alexa (Voice Assistants)
- Google Translate (Language Translation)
- Tesla Autopilot (Self-driving Cars)

What is Machine Learning (ML)?

Machine Learning is a subset of AI that focuses on enabling systems to learn from data, improve performance from experience, and make predictions or decisions without being explicitly programmed.

Basic Formula: ML = Data + Algorithms + Experience

Example: Email spam filters that learn from incoming email patterns to classify as spam or not.

What is Deep Learning (DL)

Deep Learning is a subset of ML that uses algorithms called neural networks, which are inspired by the structure and function of the brain. It is particularly effective for tasks involving large volumes of unstructured data like images, audio, and text.
Popular frameworks: TensorFlow, PyTorch

AI vs ML vs DL

The following table summarizes the difference between AI, ML, and DL:

AI: Broad area simulating human intelligence
ML: Subset of AI, focuses on learning from data
DL: Subset of ML, uses deep neural networks

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)
Definition	Simulation of human intelligence in machines	Subset of AI that enables systems to learn from data	Subset of ML using layered neural networks
Objective	To build intelligent systems that can perform tasks like humans	To develop models that can learn patterns from data	To create models that can automatically learn features from data
Human Intervention	Requires more human intervention for rule setting and updates	Less intervention needed; learns and adapts from data	Minimal human intervention, features are learned automatically
Data Dependency	Can work with both structured logic and some data	Needs structured and labelled data for training	Needs massive volumes of data, often unstructured
Hardware Requirement	Can run on basic systems	Requires moderate hardware (CPU/GPU)	Needs high-end GPUs for training deep neural networks
Interpretability	Easier to interpret and explain decisions	Moderately interpretable (e.g., decision trees)	Often a "black box" – difficult to interpret deep learning models
Examples	Expert systems, Robotics, Chatbots	Spam filters, Credit scoring, Recommendation systems	Image recognition, Natural language translation, Self-driving cars

Types of Machine Learning Algorithms

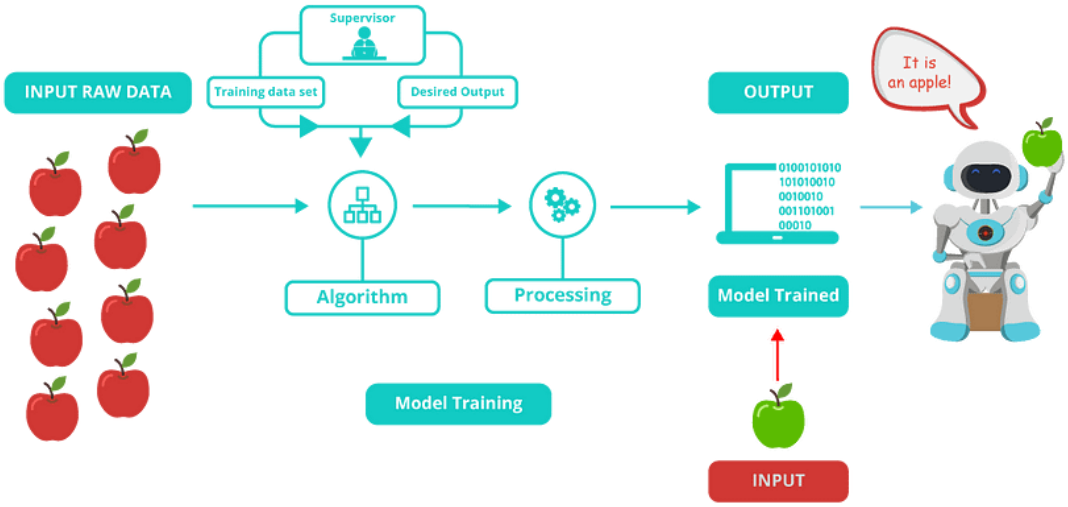
1. Supervised Learning: Trained in labeled data (e.g., classification, regression)
2. Unsupervised Learning: Finds hidden patterns in unlabeled data (e.g., clustering)
3. Reinforcement Learning: Learns optimal actions through rewards and penalties

1. Supervised Learning

- **Definition:**
  Uses labelled data to train models to make predictions or decisions. Each data point has a known output.
- Develops predictive model based on both input and output data.
- Supervised learning is the one where you have input variables (x) and an output variable (Y) and you use an algorithm to learn the mapping function from the input to the output.
  - $Y = f(X)$
- The goal is to approximate the mapping function so well that whenever you get some new input data (x), the machine can easily predict the output variables (Y) for that data.

How it works:

- Algorithm is trained using input-output pairs.
- Learns to map inputs to the correct output.
- Adjusts model based on prediction errors.



In the Supervised machine learning algorithm, every instance of the training dataset consists of input attributes and expected output. The training dataset can take any kind of data as input like values of a database row, the pixels of an image, or even an audio frequency histogram.

This category is termed as supervised learning because the process of an algorithm learning from the training dataset can be thought of as a teacher teaching his students. The algorithm continuously predicts the result based on training data and is continuously corrected by the teacher. The learning continues until the algorithm achieves an acceptable level of performance.

Applications:

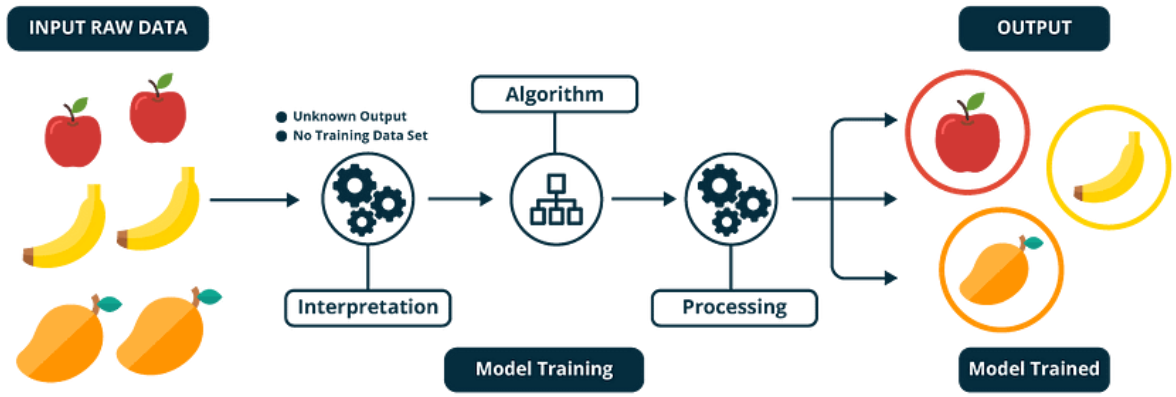
- Email spam detection
- Medical diagnosis
- Housing price prediction
- Fraud detection

2. Unsupervised Learning

- **Definition:**
  Uses un-labelled data. The model tries to find patterns, structures, or groupings within the data.
- Group and interpret data based only on input data.
- Mathematically, Unsupervised learning is where you only have input data(X) and no corresponding output variables.
- The goal for unsupervised learning is to model the underlying structure or distribution in the data to learn more about the data.

How it works:

- No target labels provided.
- Discovers hidden structure in data.



In the unsupervised learning approach, the sample of a training dataset does not have an expected output associated with them. Using the unsupervised learning algorithms, you can detect patterns based on the typical characteristics of the input data.

Clustering can be considered as an example of a machine learning task that uses the unsupervised learning approach. The machine then groups similar data samples and identifies different clusters within the data.

Applications:

- Customer segmentation
- Recommendation systems
- Anomaly detection
- Market basket analysis

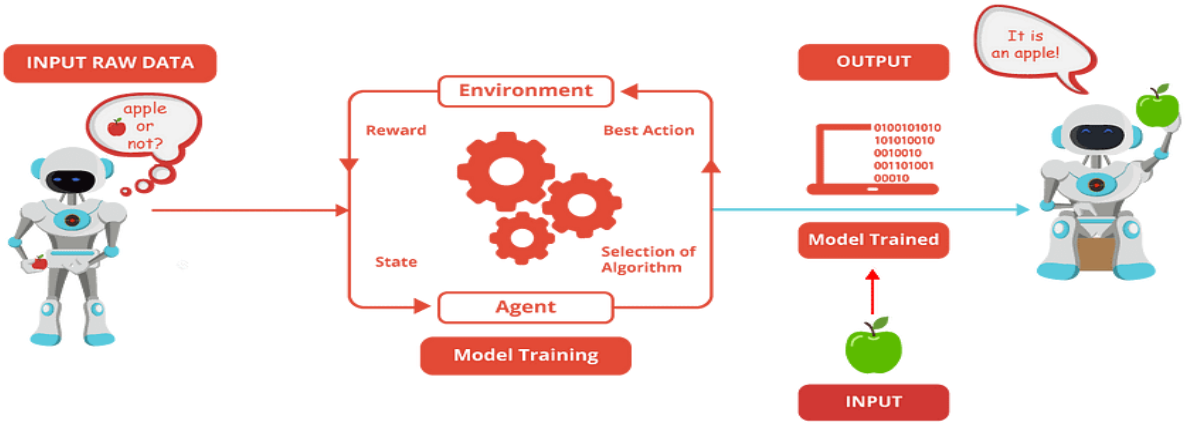
3. Reinforcement Learning

Definition:

The model (agent) interacts with an environment to learn optimal actions using trial and error and feedback via rewards.

- Reinforcement learning can be thought of as a hit and trial method of learning.
- The machine gets a Reward or Penalty point for each action it performs.

- If the option is correct, the machine gains the reward point or gets a penalty point in case of a wrong response.



How it works:

- No target labels provided.
- Discovers hidden structure in data.

The reinforcement learning algorithm is all about the interaction between the environment and the learning agent. The learning agent is based on exploration and exploitation.

Exploration is when the learning agent acts on trial and error, and Exploitation is when it performs an action based on the knowledge gained from the environment. The environment rewards the agent for every correct action, which is the reinforcement signal. With the aim of collecting more rewards obtained, the agent improves its environmental knowledge to choose or perform the next action.

Applications:

- Self-driving cars
- Robotics
- Game playing (e.g., AlphaGo)
- Stock trading

Supervised Learning vs Unsupervised Learning vs Reinforcement Learning

Aspect	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Input Data	Labelled data (features + correct output)	Unlabelled data (only features)	Agent receives input via environment states
Goal	Learn mapping from inputs to outputs	Discover hidden patterns or groupings	Learn optimal actions through trial and error
Feedback Type	Direct feedback (correct answers known)	No feedback, only structure is explored	Feedback through rewards or penalties

Aspect	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Example Use Case	Email spam detection, disease diagnosis	Customer segmentation, market basket analysis	Game playing, robot navigation
Learning Dependency	Needs large, labelled dataset	Needs large unlabelled dataset	Needs an interactive environment for learning

Applications of Machine Learning

• Healthcare: Disease prediction, diagnostics
• Finance: Credit scoring, fraud detection
• Retail: Product recommendation systems
• Social Media: Fake news detection, content moderation

Healthcare

Machine Learning is transforming the healthcare sector by enhancing diagnostic accuracy, predicting diseases, personalizing treatments, and improving operational efficiency.

Key Applications:

• *Disease Prediction and Diagnosis:*
- ML models analyze patient data to predict diseases such as diabetes, cancer, and heart disease.
- Example: Deep learning algorithms detect tumors in X-rays and MRI scans.

• *Medical Imaging:*
- CNNs (Convolutional Neural Networks) identify abnormalities in images like CT scans and pathology slides.

• *Drug Discovery and Development:*
- ML accelerates the identification of drug compounds and their predicted effects.

• *Personalized Medicine:*
- ML recommends tailored treatments based on genetics, lifestyle, and medical history.

• *Administrative Workflow Automation:*
- Automating patient scheduling, claims processing, and record management using predictive analytics.

Finance

Machine Learning in finance improves fraud detection, risk assessment, algorithmic trading, and customer service.

Key Applications:

• *Fraud Detection:*
- ML detects unusual patterns and flags fraudulent transactions in real time.
- Example: Credit card anomaly detection systems.

• *Credit Scoring and Risk Modeling:*
- Financial institutions use ML to assess the risk profile of borrowers.

• *Algorithmic Trading:*
- ML models predict stock price movements and execute high-frequency trades.

• *Customer Service Chatbots:*
- AI-powered bots handle customer queries, improve satisfaction, and reduce support costs.

• *Portfolio Management:*
- Robo-advisors recommend personalized investment strategies based on goals and risk appetite.

Retail

ML enhances the retail experience by enabling personalization, improving inventory management, and optimizing pricing.

Key Applications:

• *Recommendation Systems:*
- E-commerce platforms use ML to suggest products based on browsing history and customer behavior.

• *Demand Forecasting:*
- Predict future product demand to manage inventory efficiently.

• *Customer Segmentation:*
- ML clusters customers by preferences and behaviors for targeted marketing.

• *Dynamic Pricing:*
- Adjusting prices based on market demand, competition, and time using predictive analytics.

• *Sentiment Analysis:*
- Analyzing customer reviews to improve product offerings and services.

Social Media

ML plays a vital role in content recommendation, fake news detection, and user engagement optimization on social media platforms.

Key Applications:

• *Content Recommendation:*
- Algorithms suggest posts, videos, and advertisements tailored to individual interests.

• *Fake News and Spam Detection:*
- ML identifies misleading content and prevents it from spreading.

• *Image and Video Recognition:*
- Auto-tagging friends in photos and moderating inappropriate content using deep learning.

• *User Sentiment Analysis:*
- Understanding public opinion about topics, brands, or events using NLP techniques.

• *Influencer Analytics:*
- Identifying key influencers and analyzing their audience impact for marketing campaigns.

Historical Background and Milestones in AI

• 1950: Alan Turing proposes the Turing Test to measure machine intelligence.
• 1956: Dartmouth Conference - the birth of Artificial Intelligence as a field.
• 1986: Backpropagation algorithm popularized in neural networks.
• 2012: AlexNet wins ImageNet competition, marking breakthrough in deep learning.

Branches of AI

• Machine Learning (ML): Learning from data
• Natural Language Processing (NLP): Text and language understanding
• Computer Vision: Image and video analysis
• Expert Systems: Rule-based systems for decision making
• Robotics: Perception, control, and decision-making in robots